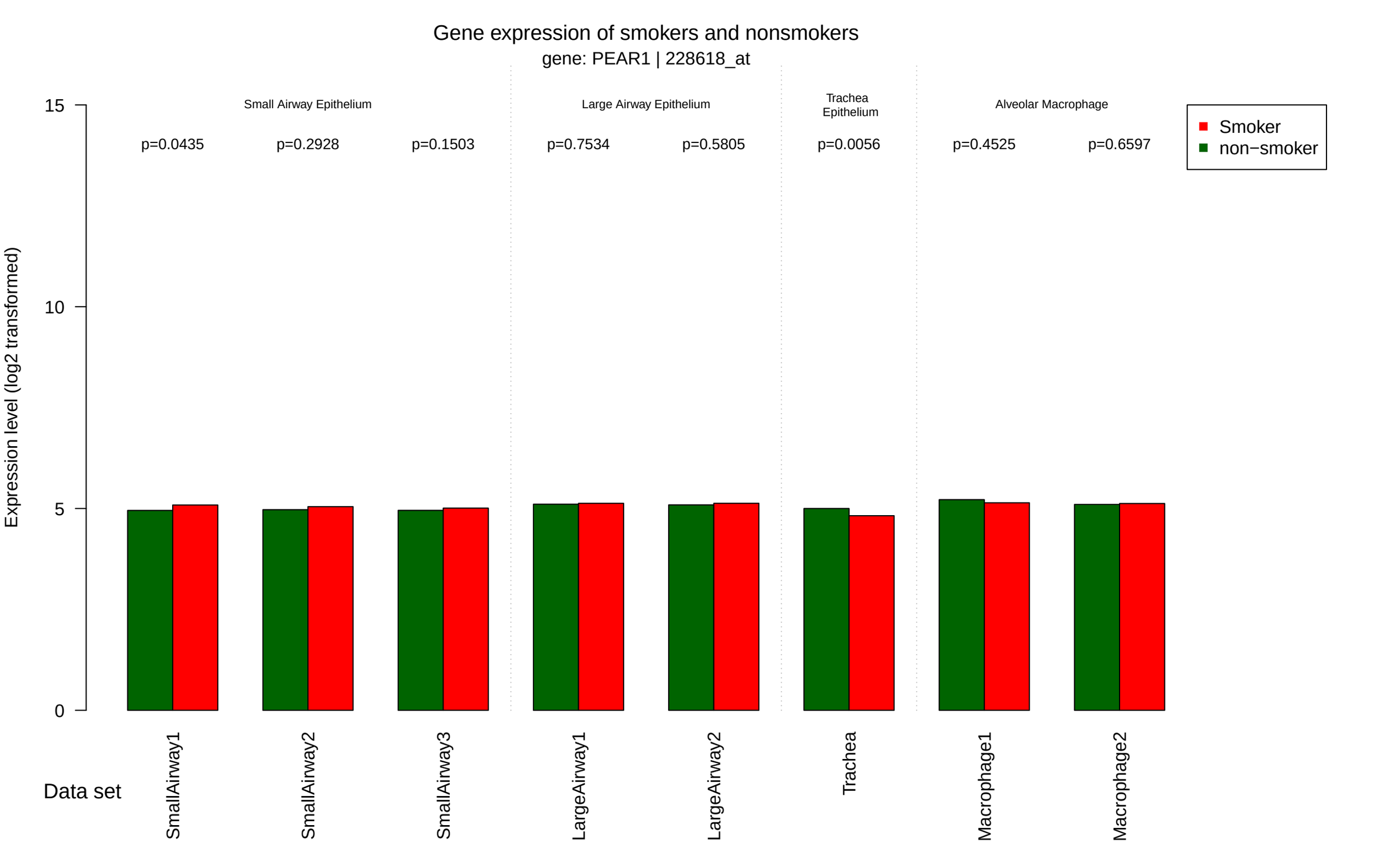
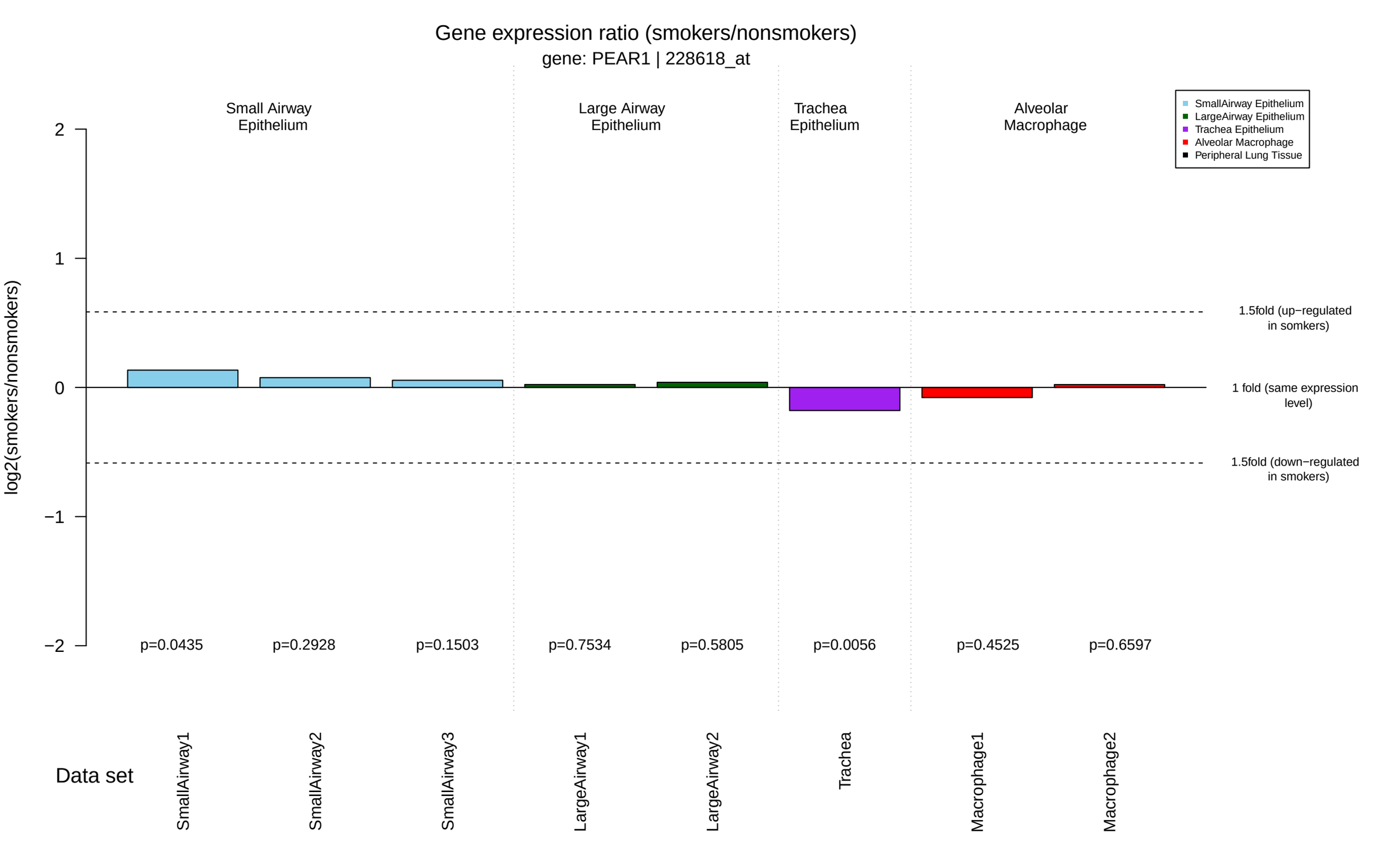
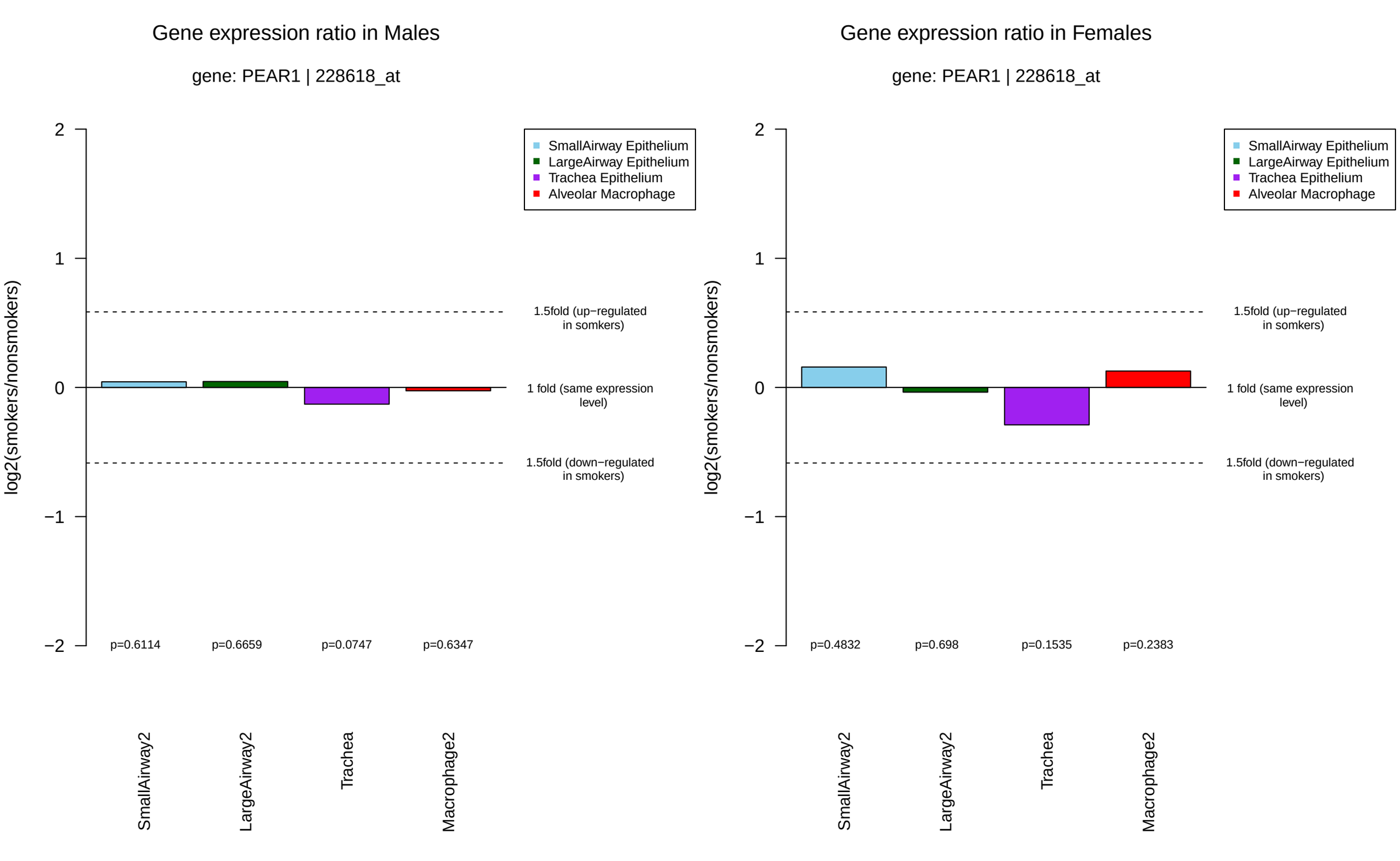
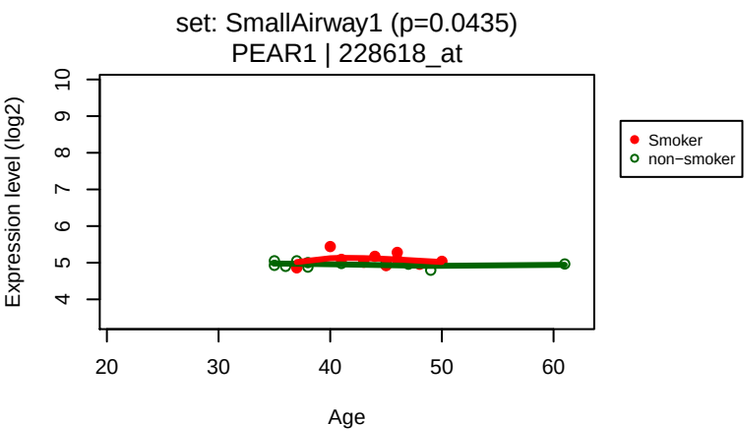
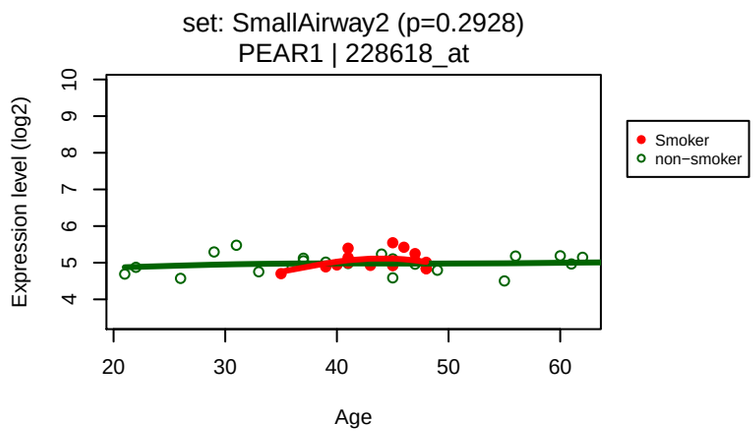
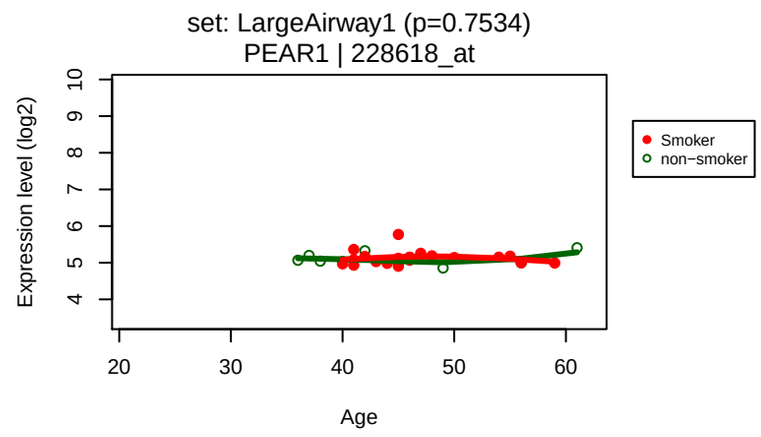

Gene expression of smokers and nonsmokers
gene: PEAR1 | 228618_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0435
p=0.2928
p=0.1503
p=0.7534
p=0.5805
p=0.0056
p=0.4525
p=0.6597
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

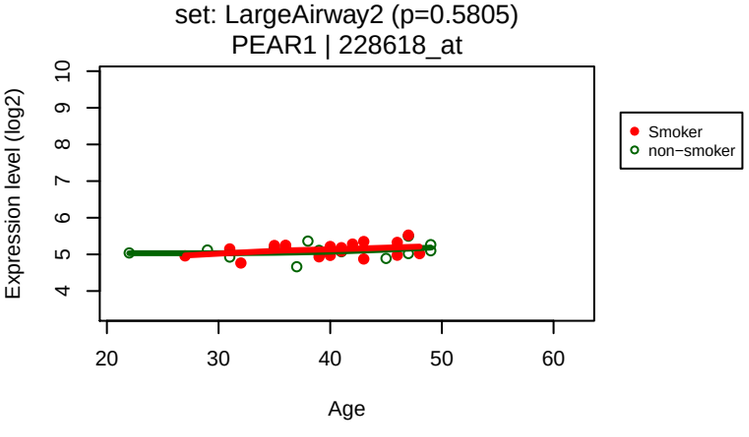

Gene expression ratio (smokers/nonsmokers)
gene: PEAR1 | 228618_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0435
p=0.2928
p=0.1503
p=0.7534
p=0.5805
p=0.0056
p=0.4525
p=0.6597
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

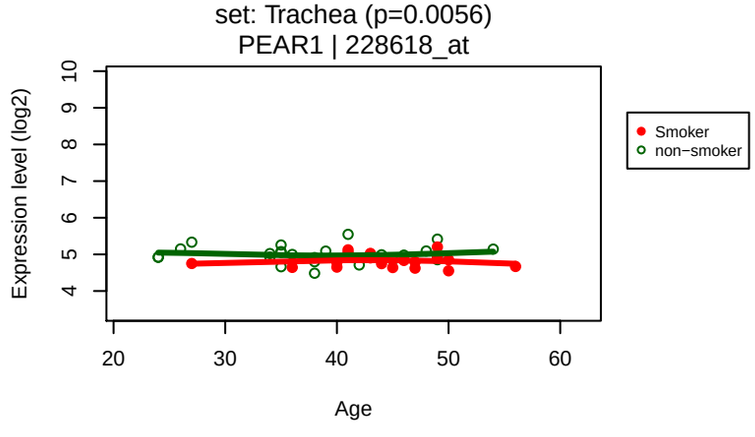

Gene expression ratio in Males
gene: PEAR1 | 228618_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6114
p=0.6659
p=0.0747
p=0.6347
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PEAR1 | 228618_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4832
p=0.698
p=0.1535
p=0.2383
SmallAirway2
LargeAirway2
Trachea
Macrophage2

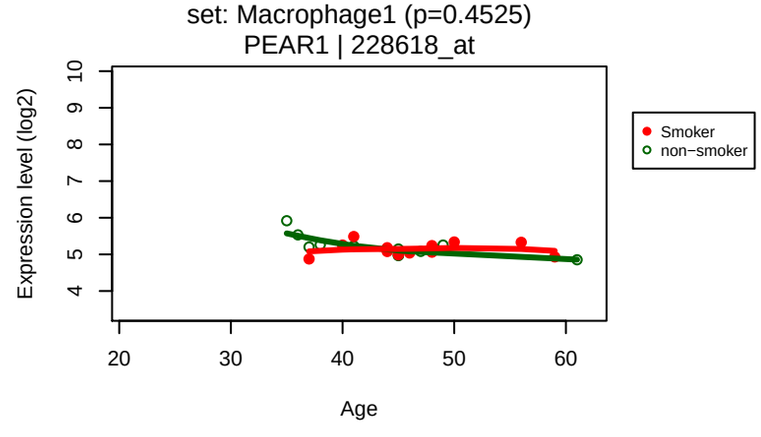

set: SmallAirway1 (p=0.0435)
PEAR1 | 228618_at
Expression level (log2)
Age
Smoker
non−smoker

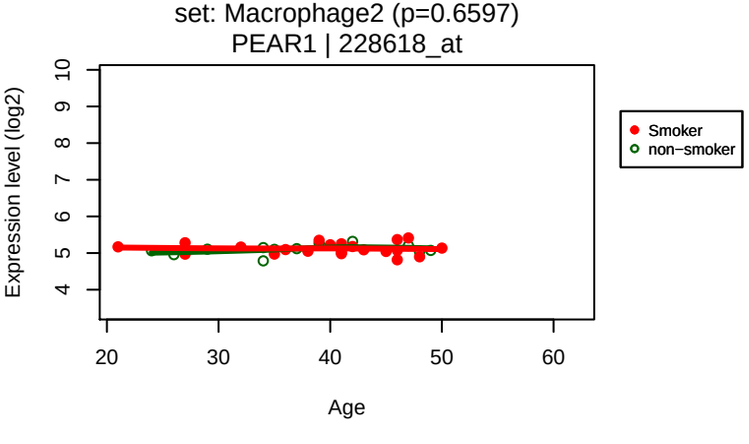

set: SmallAirway2 (p=0.2928)
PEAR1 | 228618_at
Expression level (log2)
Age
Smoker
non−smoker

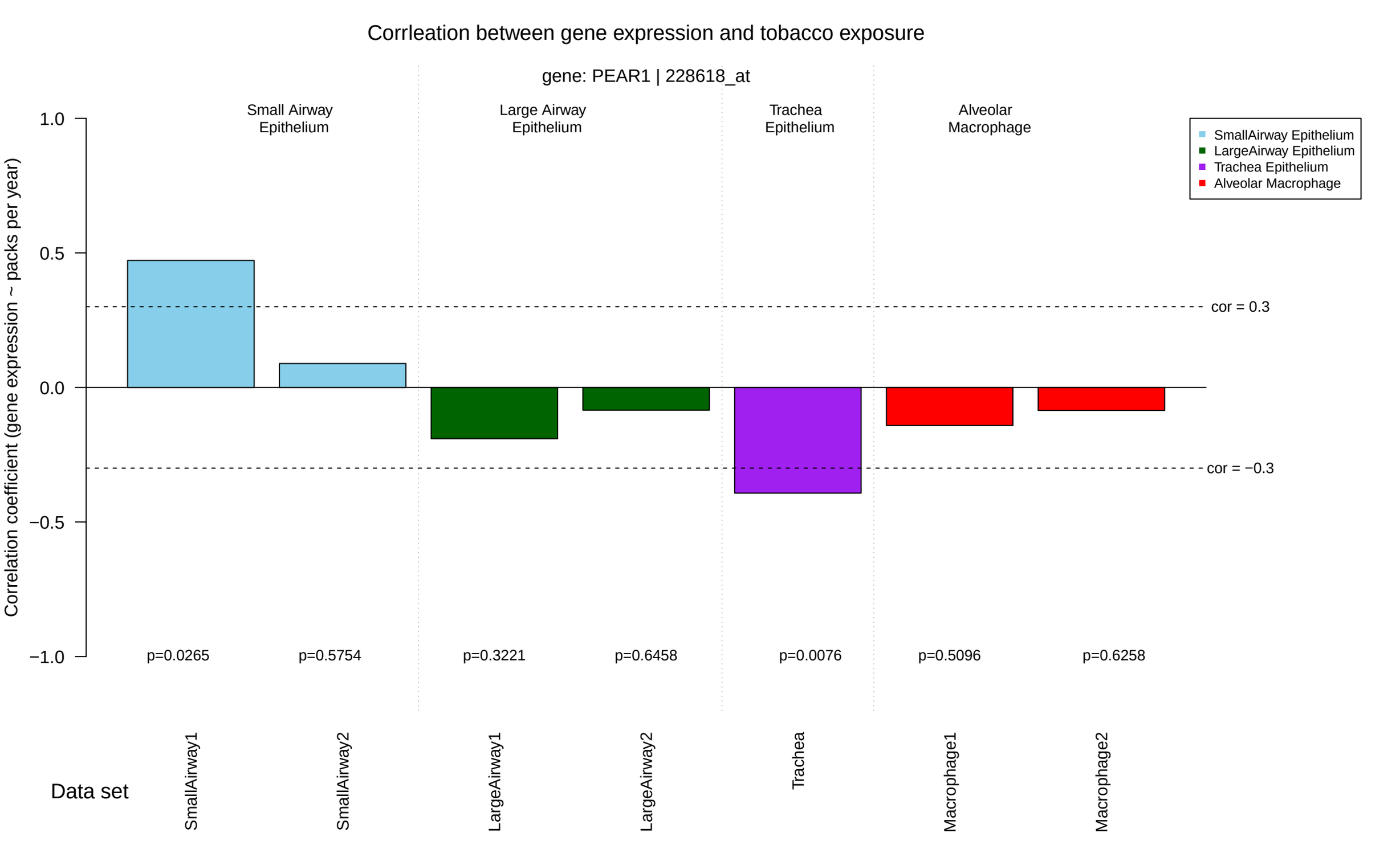

set: LargeAirway1 (p=0.7534)
PEAR1 | 228618_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5805)
PEAR1 | 228618_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0056)
PEAR1 | 228618_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4525)
PEAR1 | 228618_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6597)
PEAR1 | 228618_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PEAR1 | 228618_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0265
p=0.5754
p=0.3221
p=0.6458
p=0.0076
p=0.5096
p=0.6258
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2